Gonzalo Díaz Amor

Data Scientist | GenAI & MLOps Specialist | LLM & RAG Engineer

Email | Teléfono | LinkedIn | GitHub | Portfolio

Professional Profile

GenAI-oriented Data Scientist with hands-on experience developing ML and AI pipelines in Python, building APIs, automating data workflows, and deploying models in cloud environments. Skilled in **LLM-based architectures, embeddings, RAG pipelines, and prompt engineering**, with practical exposure to Azure, LangChain, and model operationalization. Proficient in Python, R, SQL, and cloud-based data engineering, with a proven ability to develop and implement predictive models and optimize business processes. Strong collaborative spirit and committed to continuous learning, staying at the forefront of the rapidly evolving field of data science.

Relevant Experience

AI/ML Research Engineer

BeWay | *Jan 2026 – Present*

- Implemented the **LLM-as-a-Judge** system (Gu et al., 2025) for the RNA (Reasoning Numerical Alignment) audit pipeline, designing the full architecture and evaluation taxonomy.
- Created a comprehensive prompt library and automated pipeline that performs fidelity, pairwise, and holistic-ranking checks on BOCR (Benefits, Opportunities, Costs, Risks) artifacts.
- Integrated bias-mitigation techniques (CoT (Chain of Thought) reasoning, rubric-based evaluation, position-swap voting) to ensure robust, reproducible judgments.
- Developed high-performance Python/R scripts and AWS-based tooling to run the judge at scale across multiple scenarios.
- *Tech stack:* **Python, R, AWS, Docker, Git, CI/CD**

AI/ML Specialist Data Scientist

Daiichi-Sankyo | *Oct 2025 – Dec 2025*

- Worked on a **Causal AI project** identifying key environmental factors from hundreds of data points to quantify pollution's causal effect on thousands of patient outcomes using **Double Machine Learning with Causal Forest**.
- Developed the data infrastructure and high-performance scripts for deployment on **AWS**.
- Fixed-term contract concluded due to budget restructuring; left on excellent terms with solid professional references.
- *Tech stack: Python, Google Cloud.*

Behavioral Data Scientist

BeWay | *Jun 2025 – Oct 2025*

- Supported a **Geo Data Science project** identifying the top 10 features from hundreds of data points to determine product penetration across over 60 million tickets using machine learning techniques.

- Spearheaded a **funnel optimization project** using Kaplan-Meier survival analysis on 10,000+ customers to identify conversion barriers and accelerate time-to-first-purchase.
- *Tech stack: R, Python, Quarto, AWS, PostgreSQL, Git.*

Data Science Specialist

Minsait | *Nov 2023 – Jun 2025*

- Developed a comprehensive **data lake** integrating diverse hospital data sources, ensuring accuracy and consistency for implementation in **OMOP-CDM (OHDSI)**.
- Built **RESTful APIs for ML/AI inference** (FastAPI), enabling integration of GenAI components such as embeddings, vector search, and LLM-powered analytics.
- Developed and deployed **NLP/ML models in production** using Python, including Dockerized deployments and ETL orchestration with **Apache Airflow**.
- Proficient in ETL processes using WhiteRabbit and Usagi for efficient data mapping and integration.
- *Tech stack: R, Python, Docker, FastAPI, PowerShell, GitLab, MySQL, PostgreSQL, Spark, Airflow, OHDSI.*

Data Scientist

TFS HealthScience | *Jun 2022 – May 2023*

- Acted as **programming lead** for a Phase IV study, including validation and documentation — saving the company **3+ years of research** following SDTM and ADaM standards for EMA submission.
- Applied machine learning techniques (logistic regression with regularization) to optimize predictive models, achieving a **10% performance improvement**.
- Created propensity matching between two treatments using regularized logistic regression to prevent overfitting.
- Performed sample size and power calculations in R, and authored statistical analysis plans and study reports for Phases I–IV.
- *Tech stack: R, SAS, RStudio, Python, CDISC, OHDSI.*

Data Scientist

Animal Data Analytics | *Sep 2020 – Jun 2022*

- Provided comprehensive analysis and recommended solutions for **5+ international clients** using SAS and CDISC standards.
- Applied **ML and Deep Learning algorithms** across 10+ projects, achieving at least 10% performance improvement with reports in RMarkdown.
- Supervised 1 intern and 1 data scientist in developing a regression model with a **7% performance upgrade** in R.
- Conducted **survival analysis** (Kaplan-Meier, Cox proportional hazards, log-rank tests) for time-to-event outcomes.
- *Tech stack: R, Python, SAS, RMarkdown.*

Education

- **MSc. Big Data & Data Science** | VIU – Valencia, Remote (2025)
  - Advanced training in large-scale data processing, cloud architectures, and AI/ML systems.
- **BSc. Statistics** | UVa – Valladolid, Spain (2020)
  - Strong foundation in statistical modeling, probability theory, and quantitative methods.

Notable Additional Education (see LinkedIn for more):

- **AI Agents Intensive Course** (Google x Kaggle, 2025): LLMs, agent frameworks, orchestration, evaluation, and real-world deployment workflows.
- **6.86x: Machine Learning with Python – From Linear Models to Deep Learning** (MITx, 2021): Linear models, deep learning, reinforcement learning, hands-on Python projects.
- **LangChain for LLM Development** (freeCodeCamp, 2023): Transformer architecture, embeddings, generative AI pipeline, and prompt engineering with LangChain.

Certifications

- **AI Agents Intensive Course** — Google x Kaggle *(2025)*
  - Hands-on training on designing and deploying AI agent systems using LLMs, tools, memory, and retrieval components. Covered agent frameworks, orchestration patterns, evaluation methodologies, and real-world deployment workflows.
- **LangChain for LLM Development** — freeCodeCamp *(2023)*
  - Transformer architecture, embeddings, generative AI project pipeline, and prompt engineering. LangChain as a standard interface for working with various LLMs.
- **Google Data Analytics Certificate** — Google *(2023)*
  - Comprehensive training in R, SQL, Python, and Tableau for data analysis and visualization.
- **6.86x: Machine Learning with Python – From Linear Models to Deep Learning** — MITx *(2021)*
  - In-depth introduction covering linear models, deep learning, and reinforcement learning, with hands-on Python projects.
- **Explainable AI (XAI)** — DPhi *(2023)*
  - Human-interpretable machine learning using SHAP and LIME for model explainability.
- **Biomedical Image Analysis in Python** — DataCamp *(2023)*
  - Deep learning for identification from medical images using supervised and unsupervised learning techniques.
- **Deep Learning Online Bootcamp** — DPhi *(2021)*
  - Image recognition bootcamp. Finished in the top 20% of the final competition with over 80% precision.
- **Visual Studio Code for Pharma R** — R in Pharma *(2023)*
  - Introduction to VS Code for data analysis and machine learning projects in the pharmaceutical context.
- **Data Analysis with Python** — freeCodeCamp *(2023)*
  - Fundamentals of data analysis with Python: Pandas, NumPy, Matplotlib, and data wrangling.
- **Continuous Learning in Data Science** — Various platforms (see LinkedIn) *(2020 – Present)*
  - Ongoing professional development across multiple platforms. Full list available on LinkedIn.

Technical Skills

- **Programming Languages:** Python (LLM APIs, LangChain, FastAPI), R, SQL, PowerShell.

- **GenAI & LLMs:** RAG pipelines, embeddings, vector search, prompt engineering, LangChain, LLM-based architectures.
- **ML & Data Science:** Scikit-Learn, Pandas, NumPy, Causal ML (Causal Forest, Double ML), survival analysis, predictive modeling.
- **MLOps & Cloud:** Docker, Apache Airflow, AWS, Azure, GCP, Databricks, PySpark, Hadoop.
- **Data Engineering:** ETL pipelines, FastAPI, OMOP-CDM (OHDSI), WhiteRabbit, Usagi, CDISC standards.
- **Databases & BI:** MySQL, PostgreSQL, MongoDB, Cassandra, PowerBI, Tableau.
- **Version Control & Collaboration:** Git, GitHub, GitLab.
- **Languages:** Spanish (Native), English (Advanced).

Palencia, Spain | Available for relocation/remote work.